HOTEL ROOM MANAGEMENT SYSTEM:

INTRODUCTION AND BACKGROUND:

The Hotel Room Management System is a comprehensive software application developed to optimize operations within small to medium-sized hotels. This project successfully leveraged key data structures, including Linked Lists for efficient room management, Binary Search Trees (BST) for accelerated search functionalities, and Stacks for robust undo/redo capabilities. Implemented as a Windows Forms application using C#, the system provides an intuitive graphical user interface, enabling hotel staff to effectively manage room reservations, availability, guest information, and generate vital reports. The system achieves its primary objective of streamlining hotel management processes, enhancing both efficiency and user experience.

OBJECTIVES ACHIEVED:

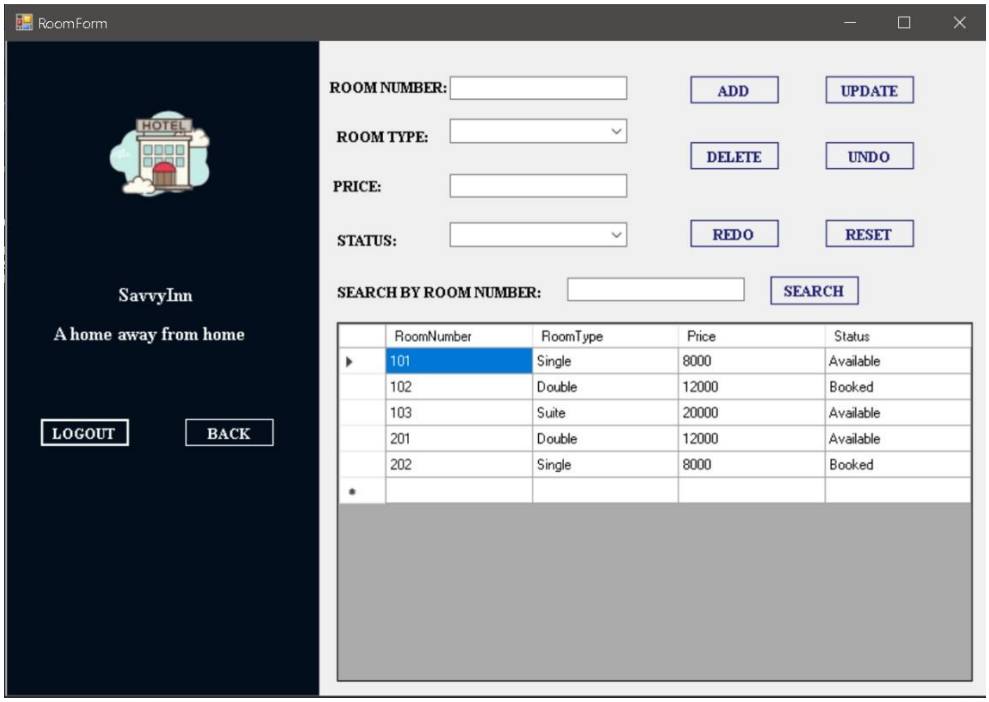
The development of the Hotel Room Management System successfully met all the predefined objectives:

- **Efficient Room Data Management:** The system successfully developed and implemented functionalities to add, update, and delete room details using Linked Lists. This allows hotel staff to maintain an up-to-date and dynamic inventory of rooms, easily reflecting changes in room status or attributes.
- **Optimized Room Search:** A Binary Search Tree (BST) was effectively integrated to provide fast and efficient room searching capabilities. The system can quickly identify available rooms based on various parameters, significantly reducing the time required for reservation inquiries.
- **Flexible Booking Modification with Undo/Redo:** Stack-based undo/redo functionality was successfully implemented for modifying booking changes. This feature greatly enhances user flexibility, allowing corrections to be made seamlessly without needing to re-enter extensive data.
- **Comprehensive Report Generation:** The system is capable of generating detailed reports on bookings, revenue, and customer statistics. These reports provide valuable insights for management, aiding in informed decision-making and strategic planning.

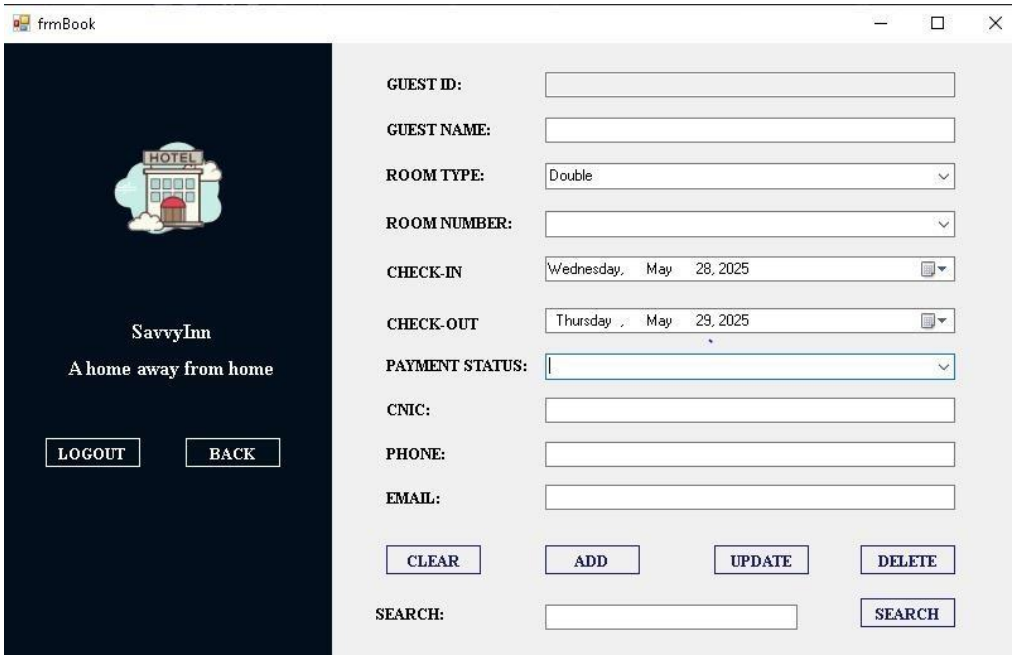
SYSTEM FEATURES AND FUNCTIONALITY:

The implemented Hotel Room Management System provides a comprehensive set of features, each meticulously designed to address specific operational needs of a hotel:

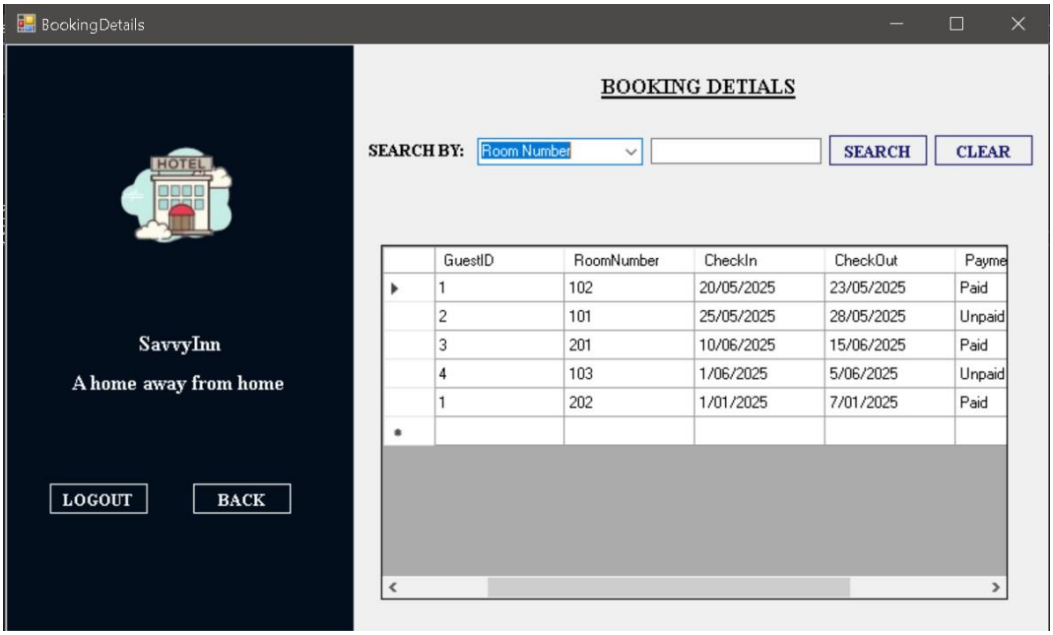
- **Room Management**: This module allows authorized hotel staff to perform full CRUD (Create, Read, Update, Delete) operations on room details. Users can add new rooms to the inventory with attributes like room number, type, price, and current status. Existing room details can be updated for changes in pricing or amenities, and rooms can be removed if no longer available. This functionality is underpinned by a robust Linked List implementation, ensuring flexible and scalable management of the hotel's room inventory. Also user can undo and redo previous operation using Stack.



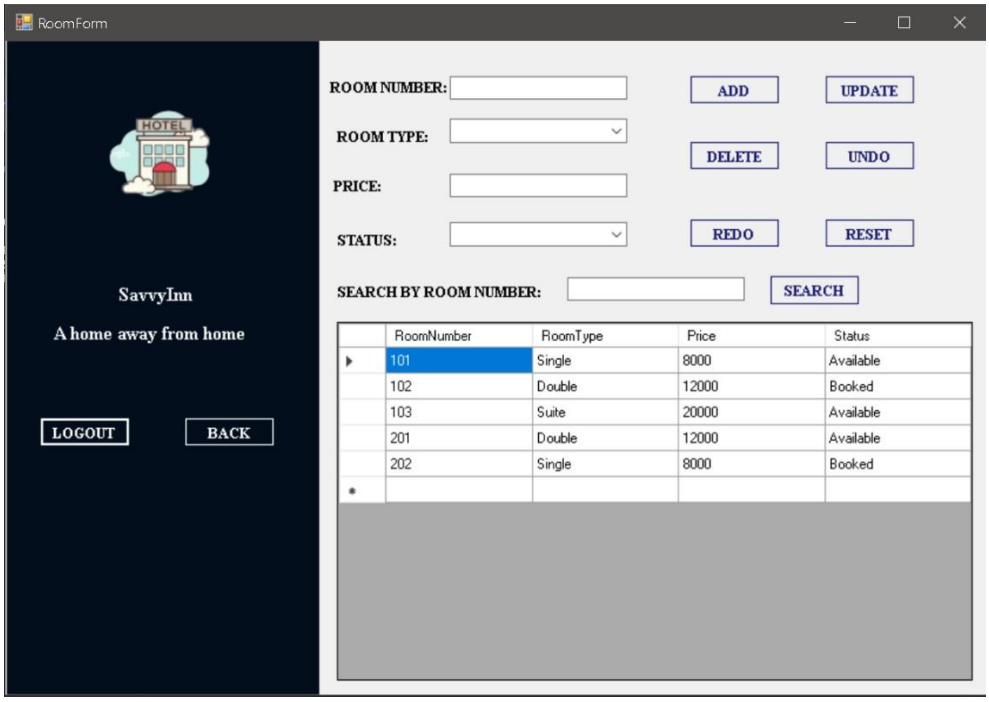
- **Booking System:** This core feature facilitates the reservation and cancellation of hotel room bookings. Staff can initiate new bookings by selecting available rooms, inputting guest details, and specifying stay durations. The system also handles cancellations, updating room availability accordingly.



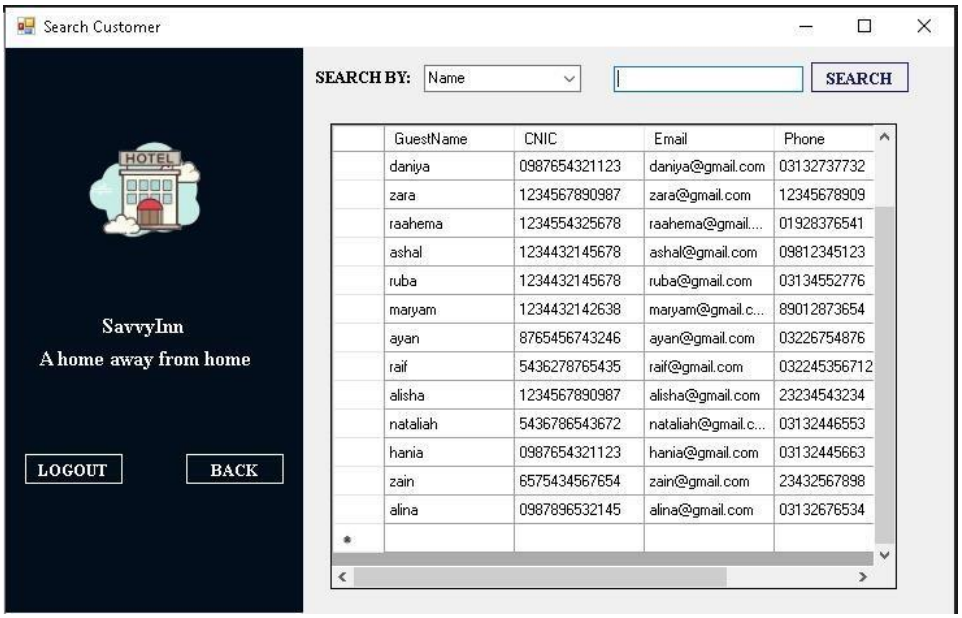
- **Search Functionality:** Leveraging a Binary Search Tree (BST), this feature enables highly efficient searching for rooms. Users can search for record based on various criteria, most notably by Room Number, but also potentially Guest Id . The BST ensures rapid retrieval of relevant room information, even with a growing number of rooms.



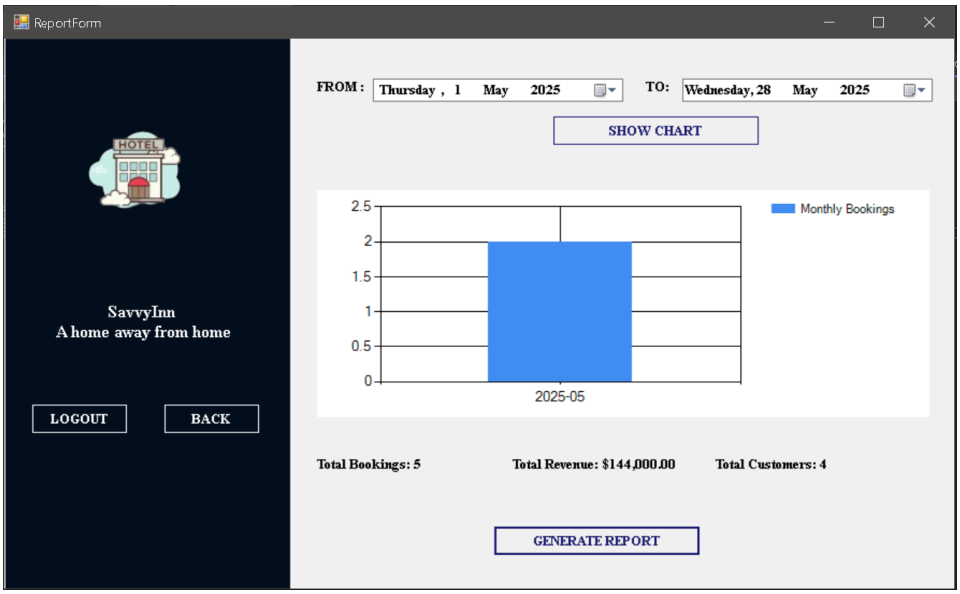
- **Undo/Redo Actions:** A crucial feature for operational flexibility and error correction, the system incorporates stack-based undo/redo functionality specifically for booking changes. If a booking modification is made incorrectly, users can easily revert the change with an "undo" action. Conversely, if an undone action needs to be reinstated, the "redo" function is available. This prevents data entry errors from causing significant rework.



- **Customer Management:** This module is dedicated to storing and managing guest details and their reservation history. It allows hotel staff to retrieve existing guest profiles by their Name or by their NIC number, and view a comprehensive record of their past bookings. This feature supports personalized service and efficient record-keeping.



- **Reports Generation:** The system offers reporting capabilities, allowing management to generate detailed insights into various aspects of hotel performance. These reports include bookings, revenue, and customer statistics.



SYSTEM ARCHITECTURE AND DESIGN:

The Hotel Room Management System follows a multi-layered architecture, typical for Windows Forms applications, to ensure modularity, maintainability, and scalability.

- **Presentation Layer (UI):** Developed using Windows Forms, this layer provides the graphical user interface that interacts directly with the hotel staff. It handles user input, displays information, and triggers business logic.
- **Business Logic Layer:** This layer contains the core logic of the application. It orchestrates the flow of data between the UI and the data access layer, implements validation rules, and manages the application's processes (e.g., booking logic, room allocation algorithms). This layer utilizes the chosen data structures (Linked Lists, BST, Stacks) for in-memory data manipulation and processing.
- **Data Access Layer (DAL):** This layer is responsible for all interactions with the database (SQL Server or SQLite). It handles data persistence, retrieval, updates, and deletions, abstracting the database specifics from the business logic.

SOFTWARE AND DEVELOPMENT ENVIRONMENT:

The project was developed using a robust set of modern software tools and technologies:

- **Programming Language:** C# was chosen for its strong object-oriented capabilities, extensive libraries, and seamless integration with the .NET ecosystem.
- **Database:** For data storage and retrieval, either SQL Server (for a robust, scalable backend) or SQLite (for a lightweight, embedded solution) was utilized, depending on the specific deployment scenario.
- **User Interface:** Windows Forms was employed for designing and implementing the graphical user interface, offering a straightforward approach to creating interactive desktop applications.
- **Core Data Structures:** Fundamental data structures such as Linked Lists, Binary Search Trees, and Stacks were implemented and utilized to ensure efficient data management and algorithm performance within the system.

CONCLUSION:

The Hotel Room Management System stands as a robust and efficient application, successfully addressing the critical needs of small to medium-sized hotels. By effectively leveraging the power of Linked Lists for dynamic room management, Binary Search Trees for optimized searching, and Stacks for flexible undo/redo functionalities, the system provides a comprehensive and user-friendly solution for hotel operations. The project successfully met all its objectives, delivering a functional and reliable system that enhances efficiency in room management, booking processes, and customer handling. This system is a valuable asset, designed to streamline daily tasks, reduce manual errors, and provide actionable insights through detailed reporting, ultimately contributing to improved guest satisfaction and operational productivity